

Cricket Match Performance Prediction using Data Analysis, Machine Learning, and MySQL

1. Abstract

This project focuses on analyzing cricket match performance data and predicting two important outcomes: total runs scored and number of wickets taken. The dataset containing cricket match statistics was analysed using Python. Exploratory Data Analysis (EDA) was performed to understand patterns and relationships between variables. Machine Learning models were then developed using the Random Forest algorithm to predict runs and wickets. The predictions generated by the model were stored in a MySQL database, creating a complete pipeline from data analysis to database storage. This project demonstrates how data analytics, machine learning, and database systems can be integrated for sports performance prediction.

2. Introduction

Cricket is one of the most data-driven sports, where statistics such as runs, wickets, strike rate, and economy rate play a crucial role in evaluating player performance. With the advancement of data science, predictive analytics can be used to forecast match outcomes and player contributions.

In this project, historical cricket data is analysed to build predictive models. The project uses Python for data analysis and machine learning, and MySQL for storing prediction results. The integration of these technologies helps create a real-world data analytics workflow.

3. Goal of the Project

The main goals of this project are:

- To analyze cricket match performance data.

- To identify relationships between batting and bowling statistics.

- To build machine learning models that predict runs and wickets.

- To store prediction results in a MySQL database.

- To demonstrate an end-to-end data analytics pipeline.

4. Dataset Overview

The dataset used in this project contains 300 records of cricket performance data. Each record represents match statistics such as runs scored, number of boundaries, overs bowled, and wickets taken.

Dataset Features

These variables are used to train machine learning models.

5. Data Preprocessing

Before building machine learning models, the dataset was cleaned and prepared.

Steps performed:

- Loading the dataset using Pandas.

- Selecting only the required columns.

Handling missing values using fillna(0).

Ensuring all numerical fields are valid for analysis.

Preparing feature variables and target variables.

Preprocessing helps improve model accuracy and reliability.

6. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure of the dataset and relationships between variables.

Visualizations Created

Distribution of Runs Histogram showing how runs are distributed across matches.

Distribution of Wickets Shows how frequently different wicket counts occur.

Overs vs Wickets Scatter Plot Analyses how bowling overs affect wicket-taking ability.

Economy vs Wickets Relationship Evaluates whether bowlers with lower economy rates take more wickets.

Boundary Runs vs Run Rate Demonstrates how boundaries influence scoring rate.

Batting Order vs Runs Shows how batting position affects total runs scored.

EDA helps identify patterns and trends within the dataset.

7. Machine Learning Model

Algorithm Used

The project uses the Random Forest Regressor algorithm.

Reasons for choosing Random Forest:

Works well with numerical datasets

Handles non-linear relationships

Reduces overfitting

Provides accurate predictions

Model 1: Runs Prediction

Target Variable

Runs scored

Features Used

Batting order

Number of fours

Number of sixes

Dot balls

Run rate

The model predicts the expected number of runs scored based on batting performance.

Model 2: Wickets Prediction

Target Variable

Wickets taken

Features Used

Overs bowled

Runs conceded

Economy rate

Maiden overs

Wides

No balls

This model predicts the number of wickets based on bowling performance.

8. Model Evaluation

The dataset was divided using train-test split:

80% training data

20% testing data

This resulted in:

240 training rows

60 testing rows

Predictions were generated for the testing dataset and evaluated using the R^2 score.

The R^2 score measures how well the predicted values match the actual values.

Higher R^2 values indicate better model performance.

9. MySQL Database Integration

To store prediction results, the project integrates Python with MySQL database.

Database Used

cricket_db

Table Created

ball_predictions

The Python program inserts prediction results into MySQL using mysql-connector-python.

Example stored data:

A total of 60 prediction rows were inserted into the database.

10. Project Workflow

Dataset CSV ↓ Python Data Cleaning ↓ Exploratory Data Analysis ↓ Machine Learning Model Training ↓ Prediction Generation ↓ Store Predictions in MySQL Database

This workflow demonstrates a real-world data analytics pipeline.

11. Tools and Technologies Used

12. Results

The machine learning model successfully predicted runs and wickets based on batting and bowling statistics. The predictions were stored in a MySQL database for further analysis.

The project demonstrates that machine learning models can effectively analyse sports performance data and generate meaningful predictions.

13. Conclusion

This project successfully integrates data analysis, machine learning, and database systems to predict cricket performance metrics. By analyzing match statistics and applying the Random Forest algorithm, the system can estimate runs scored and wickets taken.

The integration with MySQL allows prediction results to be stored and used for further visualization or dashboard creation. This project demonstrates the potential of data analytics in sports performance evaluation.

14. Future Enhancements

The project can be further improved by:

Increasing dataset size for better accuracy.

Adding player or match identifiers.

Creating Power BI dashboards connected to MySQL.

Using advanced models such as Gradient Boosting or XGBoost.

Developing a real-time prediction system.

15. References

[Scikit-learn Documentation](#)

[Pandas Documentation](#)

[Matplotlib Visualization Guide](#)

[MySQL Official Documentation](#)